INORGANIC CUMULATIVE EXAM

10-16-15

1) NH_3 and BH_3 have two different structures.

(a) Draw the structures for the monomeric forms of NH_3 and BH_3. Be sure to include perspective drawings where needed (hashes and wedges).

(b) To what point groups do these molecules belong?

Differences in the structures of NH_3 and BH_3 can be determined from the gas phase vibrational spectra of the monomers of each molecule.

2) Use the group theory method to determine how many IR and Raman active normal modes would be observed for NH_3 (show all your work)

3) The reducible representation for all of the normal modes of BH_3 is $\Gamma_{3N} = A'_1 + A'_2 + 3E' + A''_1 + A''_2 + E''$ How many IR and Raman active modes are expected for a BH_3 monomer? Which irreducible representations correspond to these normal modes of vibration?

Trimethyl tantalum has a trigonal pyramidal structure in the gas phase. A perspective drawing of Trimethyl tantalum is drawn below.

4) Using group theory, determine what orbitals on Ta are of appropriate symmetry to form Ta-Methyl sigma bonds? (Show all of your work)

5) Which orbitals on Ta are likely of the appropriate energy to form Ta-Methyl sigma bonds?

6) How many ^1H NMR signals would you expect for trimethyl tantalum? Assume there is **no** J coupling between Ta and H.

Some Useful information

$$\# reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$

$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$

$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$

$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$

$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		$(x^2-y^2$, xy)

C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	Rz	
E_1	2	$2\cos(2\pi/5)$	$2\cos(4\pi/5)$	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	$2\cos(4\pi/5)$	$2\cos(2\pi/5)$	0		(x^2-y_2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	